

回归分析 上机HW3 2112054 王赫明



3.对文件“5-7.xlsx”中的数据考虑如下方法选择子集回归模型:

```
In [4]: #读取数据
data=read.csv("lrhw3.csv")
```

(1) 用向前法建立子集回归模型;

```
In [7]: #从只含截距项的模型出发
lm.forward_0<-lm(y~1, data=data)

#向前法回归
lm.forward<-step(lm.forward_0, scope=list (upper=~x1+x2+x3+x4+x5, lower=~1), direction="forward")

summary(lm.forward)
```

Start: AIC=182.59

$y \sim 1$

	Df	Sum of Sq	RSS	AIC
+ x4	1	10578.7	4103.2	147.62
+ x1	1	5783.8	8898.1	170.06
+ x5	1	1784.4	12897.6	180.83
<none>			14681.9	182.59
+ x3	1	184.5	14497.5	184.22
+ x2	1	153.8	14528.1	184.28

Step: AIC=147.61

$y \sim x4$

	Df	Sum of Sq	RSS	AIC
+ x3	1	2028.91	2074.3	129.83
+ x5	1	1473.88	2629.4	136.71
<none>			4103.2	147.62
+ x1	1	195.81	3907.4	148.20
+ x2	1	0.12	4103.1	149.61

Step: AIC=129.83

$y \sim x4 + x3$

	Df	Sum of Sq	RSS	AIC
+ x2	1	226.269	1848.1	128.48
<none>			2074.3	129.83
+ x1	1	89.172	1985.2	130.56
+ x5	1	77.805	1996.5	130.72

Step: AIC=128.48

$y \sim x4 + x3 + x2$

	Df	Sum of Sq	RSS	AIC
+ x1	1	246.163	1601.9	126.34
<none>			1848.1	128.48
+ x5	1	12.944	1835.1	130.28

Step: AIC=126.34

$y \sim x4 + x3 + x2 + x1$

	Df	Sum of Sq	RSS	AIC
+ x5	1	120.85	1481.0	126.06
<none>			1601.9	126.34

Step: AIC=126.06

$y \sim x4 + x3 + x2 + x1 + x5$

Call:

```
lm(formula = y ~ x4 + x3 + x2 + x1 + x5, data = data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-13.7049	-2.8200	0.6207	3.9062	17.3937

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	326.37699	95.85338	3.405	0.00243	**
x4	-22.97522	2.69281	-8.532	1.4e-08	***
x3	3.77078	1.45520	2.591	0.01633	*
x2	2.54722	1.24523	2.046	0.05240	.
x1	0.06780	0.02891	2.345	0.02803	*
x5	2.45572	1.79259	1.370	0.18394	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.025 on 23 degrees of freedom

Multiple R-squared: 0.8991, Adjusted R-squared: 0.8772

F-statistic: 41 on 5 and 23 DF, p-value: 1.034e-10

依次引入 x_4 , x_3 , x_2 , x_1 , x_5 , 五个自变量, 但可以发现, x_2 与 x_5 的p值较大, 二者的显著性较差。

(2) 用逐步回归法建立子集回归模型:

```
In [8]: lm.step_0<-lm(y~., data=data)

#向前法回归
lm.step<-step(lm.step_0, scope=list (upper=~x1+x2+x3+x4+x5, lower=~1), direction ="f

summary(lm.step)
```

Start: AIC=126.06

$y \sim x_1 + x_2 + x_3 + x_4 + x_5$

Call:

`lm(formula = y ~ x1 + x2 + x3 + x4 + x5, data = data)`

Residuals:

	Min	1Q	Median	3Q	Max
	-13.7049	-2.8200	0.6207	3.9062	17.3937

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	326.37699	95.85338	3.405	0.00243	**
x1	0.06780	0.02891	2.345	0.02803	*
x2	2.54722	1.24523	2.046	0.05240	.
x3	3.77078	1.45520	2.591	0.01633	*
x4	-22.97522	2.69281	-8.532	1.4e-08	***
x5	2.45572	1.79259	1.370	0.18394	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.025 on 23 degrees of freedom

Multiple R-squared: 0.8991, Adjusted R-squared: 0.8772

F-statistic: 41 on 5 and 23 DF, p-value: 1.034e-10

结果与向前法基本一致，引入了全部自变量， x_2 与 x_5 的p值与其余自变量相比仍然较大。

考虑手动剔除不显著变量，修正回归模型。（以逐步回归为例，二者实质上等价）从AIC最小的 x_5 入手：

```
In [13]: lm_1234<-lm(y~x1+x2+x3+x4, data=data)
summary(lm_1234)
```

Call:

```
lm(formula = y ~ x1 + x2 + x3 + x4, data = data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-14.322	-2.639	0.025	4.786	16.003

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	270.21013	88.21060	3.063	0.00534 **
x1	0.05156	0.02685	1.920	0.06676 .
x2	2.95141	1.23167	2.396	0.02471 *
x3	5.33861	0.91506	5.834	5.13e-06 ***
x4	-21.11940	2.36936	-8.914	4.42e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.17 on 24 degrees of freedom

Multiple R-squared: 0.8909, Adjusted R-squared: 0.8727

F-statistic: 48.99 on 4 and 24 DF, p-value: 3.327e-11

再剔除p值最大，显著性最差的 x_1 :

```
In [14]: lm_234<-lm(y~x2+x3+x4, data=data)
summary(lm_234)
```

Call:

```
lm(formula = y ~ x2 + x3 + x4, data = data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-13.9748	-3.9549	-0.0943	3.8171	20.5014

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	389.1659	66.0937	5.888	3.83e-06 ***
x2	2.1247	1.2145	1.750	0.0925 .
x3	5.3185	0.9629	5.523	9.69e-06 ***
x4	-24.1324	1.8685	-12.915	1.46e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.598 on 25 degrees of freedom

Multiple R-squared: 0.8741, Adjusted R-squared: 0.859

F-statistic: 57.87 on 3 and 25 DF, p-value: 2.167e-11

仅剩下 x_2 一个p值较大的变量，最后剔除 x_2 :

```
In [21]: lm_34<-lm(y~x3+x4, data=data)
summary(lm_34)
```

Call:

```
lm(formula = y ~ x3 + x4, data = data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-15.2773	-3.5636	-0.9829	4.8435	23.5852

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	483.6703	39.5671	12.224	2.78e-12 ***
x3	4.7963	0.9511	5.043	3.00e-05 ***
x4	-24.2150	1.9405	-12.479	1.75e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.932 on 26 degrees of freedom

Multiple R-squared: 0.8587, Adjusted R-squared: 0.8478

F-statistic: 79.01 on 2 and 26 DF, p-value: 8.938e-12

剩余的 $x_{\{3\}}$, $x_{\{4\}}$ 的p值均较小, 可认为筛选剔除后的自变量是显著的!

此时, 回归方程为: $\hat{y} = 483.6703 + 4.7963x_3 - 24.2150x_4$.

(3)应用所有可能子集回归法建立子集回归模型, 计算 RMS_q 和AIC值, 你推荐哪一个子集回归模型, 为什么?

```
In [78]: library(leaps)
expm <- regsubsets(y ~ x1+x2+x3+x4+x5, data=data, nbest=31)
sum <- summary(expm)
sum
```

Subset selection object

Call: regsubsets.formula(y ~ x1 + x2 + x3 + x4 + x5, data = data, nbest = 31)

5 Variables (and intercept)

Forced in Forced out

x1	FALSE	FALSE
x2	FALSE	FALSE
x3	FALSE	FALSE
x4	FALSE	FALSE
x5	FALSE	FALSE

31 subsets of each size up to 5

Selection Algorithm: exhaustive

		x1	x2	x3	x4	x5
1	(1)	" "	" "	" "	" "	"*"
1	(2)	"*"	" "	" "	" "	" "
1	(3)	" "	" "	" "	" "	"*"
1	(4)	" "	" "	" "	"*"	" "
1	(5)	" "	"*"	" "	" "	" "
2	(1)	" "	" "	" "	"*"	"*"
2	(2)	" "	" "	" "	" "	"*"
2	(3)	"*"	" "	" "	" "	"*"
2	(4)	" "	"*"	" "	" "	"*"
2	(5)	"*"	"*"	" "	" "	" "
2	(6)	"*"	" "	"*"	" "	" "
2	(7)	"*"	" "	" "	" "	"*"
2	(8)	" "	" "	"*"	" "	"*"
2	(9)	" "	"*"	" "	" "	"*"
2	(10)	" "	"*"	"*"	" "	" "
3	(1)	" "	"*"	"*"	"*"	" "
3	(2)	"*"	" "	"*"	"*"	" "
3	(3)	" "	" "	"*"	"*"	"*"
3	(4)	"*"	" "	" "	"*"	"*"
3	(5)	" "	"*"	" "	"*"	"*"
3	(6)	"*"	"*"	" "	"*"	" "
3	(7)	"*"	"*"	"*"	" "	" "
3	(8)	"*"	" "	"*"	" "	"*"
3	(9)	" "	"*"	"*"	" "	"*"
3	(10)	"*"	"*"	" "	" "	"*"
4	(1)	"*"	"*"	"*"	"*"	" "
4	(2)	"*"	" "	"*"	"*"	"*"
4	(3)	" "	"*"	"*"	"*"	"*"
4	(4)	"*"	"*"	" "	"*"	"*"
4	(5)	"*"	"*"	"*"	" "	"*"
5	(1)	"*"	"*"	"*"	"*"	"*"

```
In [79]: rms=sqrt(sum$rss/29)
rms
```

```
11.8950057662516 · 17.5166404543341 · 21.0889443344717 ·
22.3587193175933 · 22.3823523762887 · 8.45745961925197 ·
9.52195158081538 · 11.6077241516037 · 11.8948383951153 ·
16.6985782601175 · 17.0416106131614 · 17.5104340333705 ·
17.9089470644659 · 21.0128110078232 · 22.1115075251818 ·
7.98287293298529 · 8.27367654856892 · 8.2973297842292 · 8.31280184666156 ·
9.52163641838663 · 11.5576171071288 · 15.4305202357692 ·
16.4681380738319 · 16.5939132540665 · 16.6077103045364 ·
7.43221937071359 · 7.7693000578946 · 7.95486709476765 · 8.12281728502132 ·
14.5846627836581 · 7.14637806824436
```

```
In [80]: #按照最优子集回归模型评价标准找到最佳组合
which.min(sum$cp)#马洛斯CP最小值
which.max(sum$adjr2) #调整R2最大值
which.min(sum$bic) #贝叶斯信息准则最小值
```

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按照 R^2 准则, x_1 - x_5 全部选入;

按照 C_p 准则与BIC准则, 删除 x_5 , 保留 x_1 - x_4